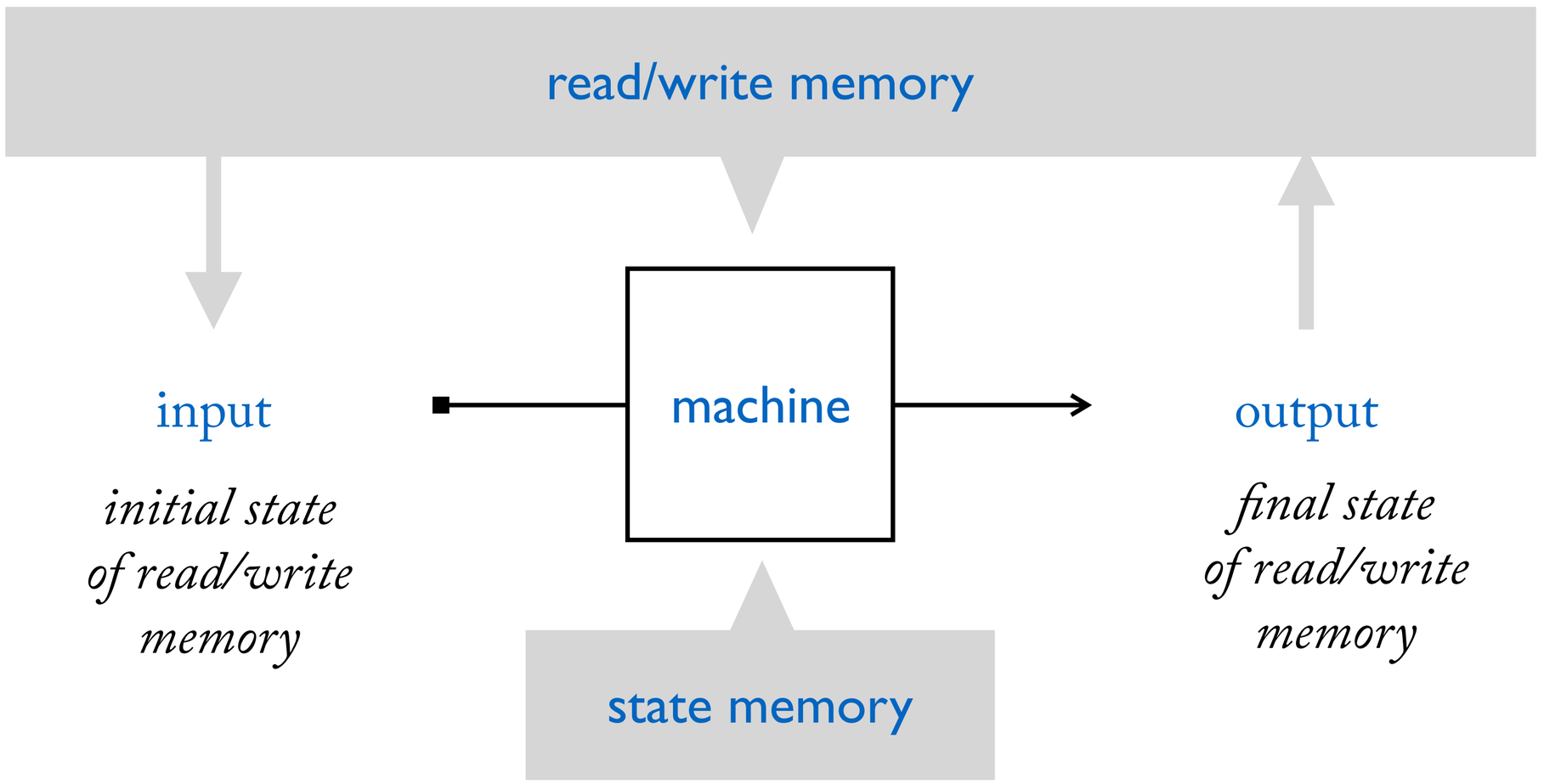
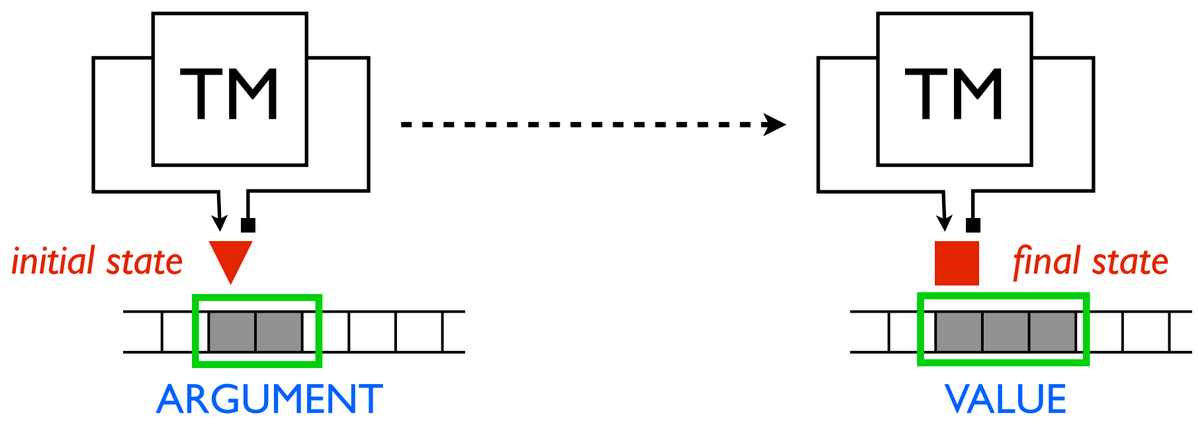
28. Computing with Turing Machines

Computing with Read/Write Memory

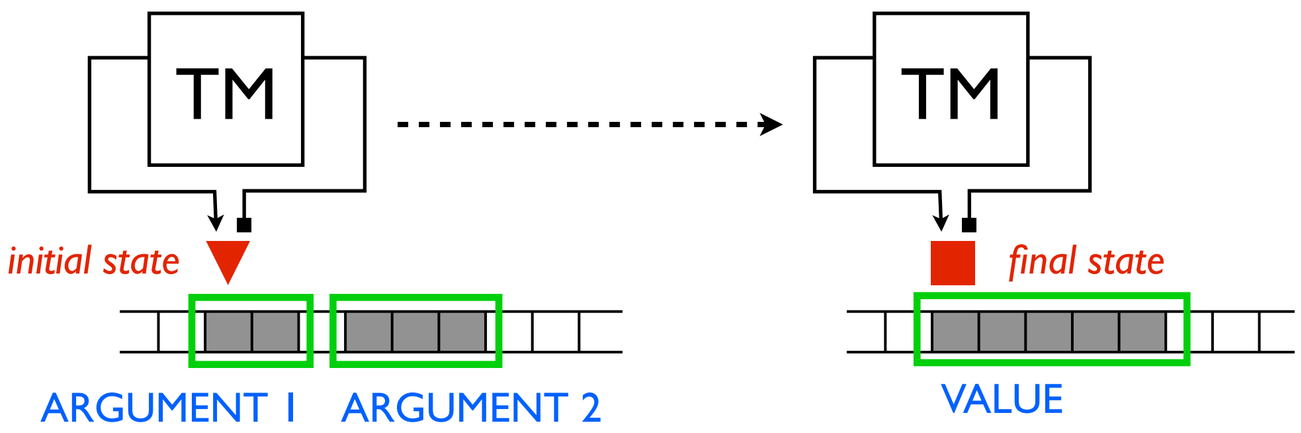
For TMs, the read/write memory serves a dual role. It functions as the input and output of the machine. But it also functions as "scratch paper" that the machine uses in the course of executing its algorithm.



Computing a one place function:

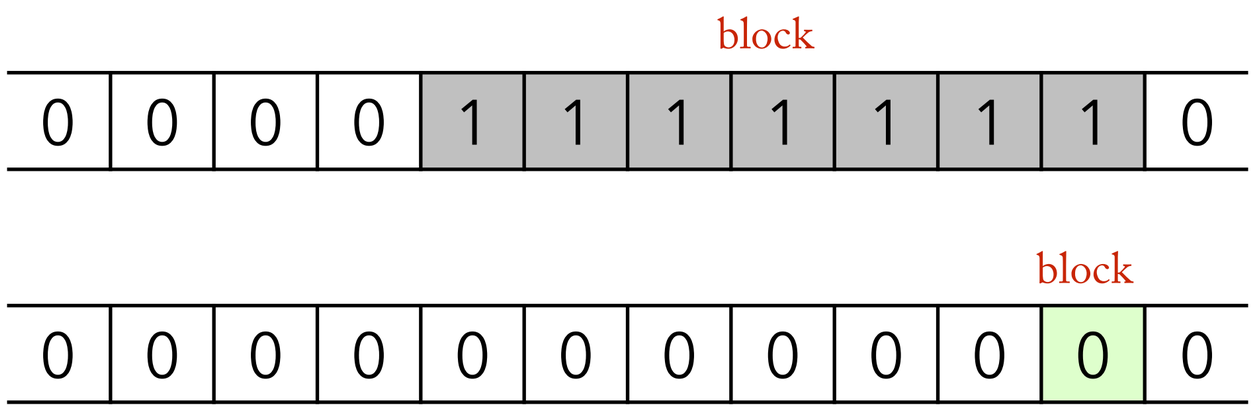


Computing a two place function:



Definition of a tally block

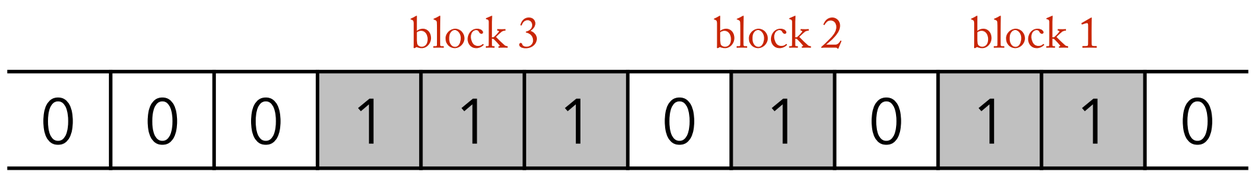
A tally block is either (a) one or more consecutive 1's; or (b) a single empty cell, that serve to represent the argument to a function.



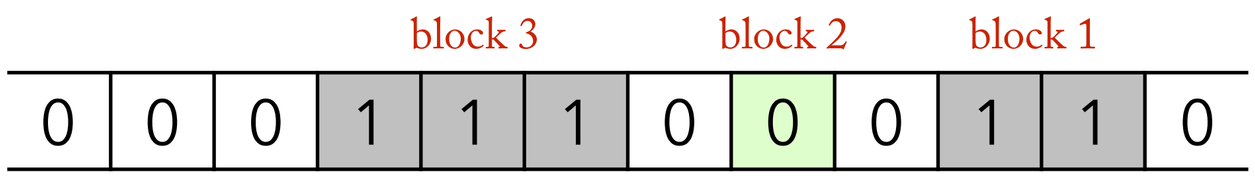
Definition of a tally input

The initial state of a tape counts as **an input** for computing a function *f* iff:

Condition 1: the tape contains a series of blocks, each separated by a single 0.



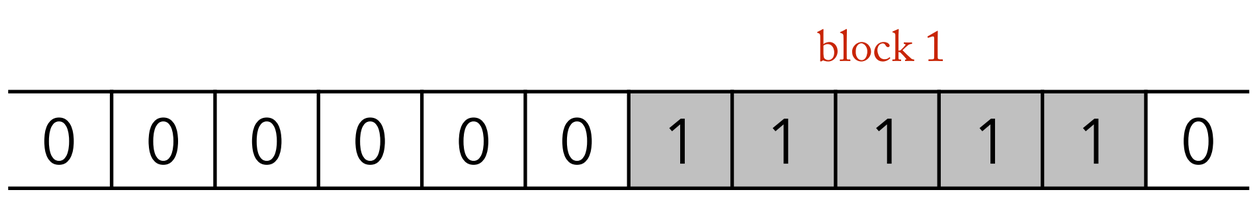
If a block is empty, it still takes up one cell.



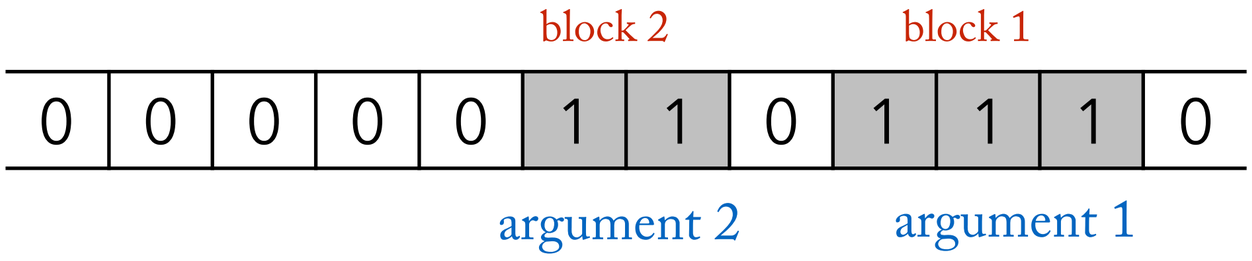
The initial blocks may be located anywhere on the tape.

Condition 2: there are as many blocks on the tape as there are arguments for the function being computed.

$f(x) = x + 1$



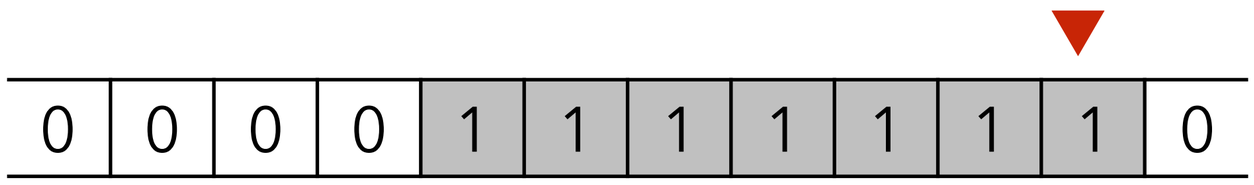
$f(x,y) = x + y$



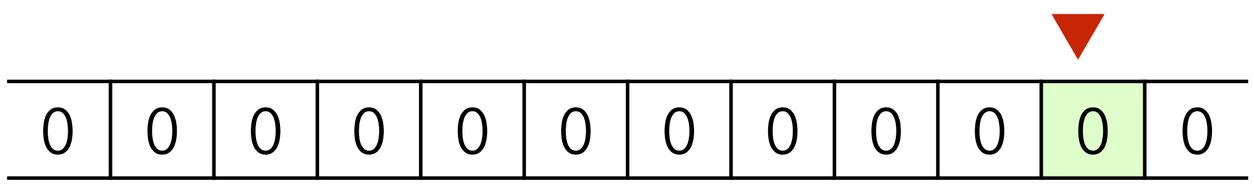
Counting from the right, the first block on the tape corresponds to the first argument of the function computed, the second block corresponds to the second argument, and so on.

Note: if the machine in question is a sub-routine of a larger machine, there may be other blocks on the tape that are irrelevant to its operation.

Condition 3: the machine starts in **standard position.** For tally, standard position is defined as **the right most cell of the right most block**.



Even if the block is empty.



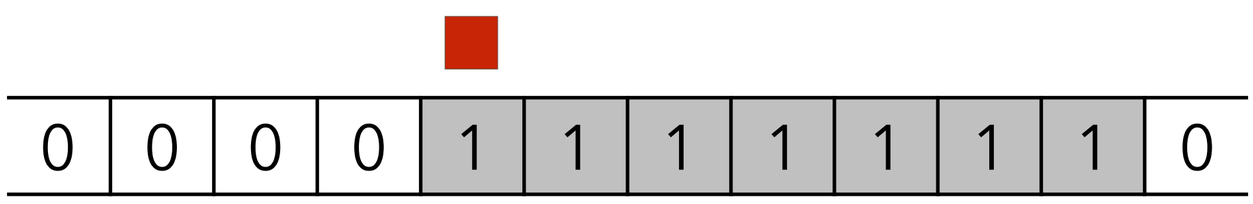
Note: If all blocks are empty, it doesn't matter where on the tape the machine starts.

Definition of a tally output

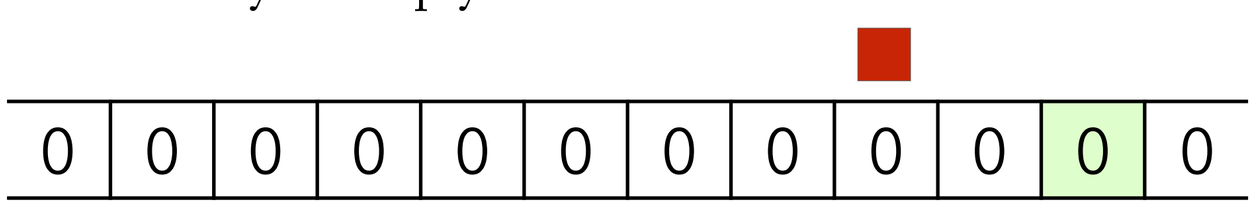
The initial state of a tape counts as **an output** for computing a function *f* iff:

Condition 1: the machine halts at some point.

Condition 2: when the machine halts, there is only a single block left on the tape.



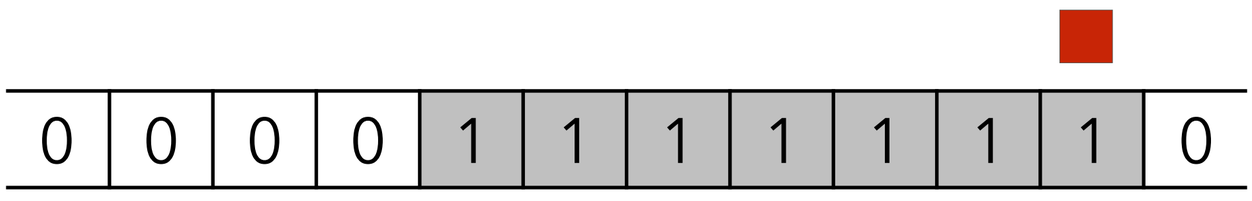
The final block may be empty:



The final block may be located anywhere on the tape.

In some circumstances, a third condition will be added. You can assume it does *not* apply, unless explicitly required.

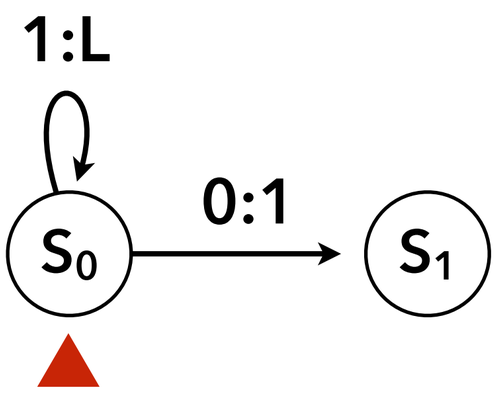
(Condition 3) The machine halts in standard position.



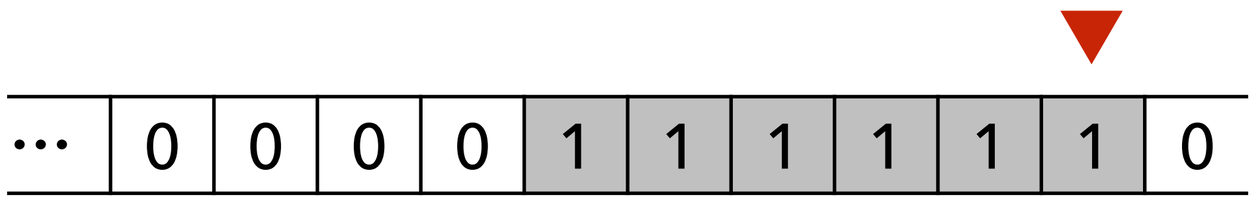
Note: standard position here is the right most cell of the right most block. It is *not* necessarily the same standard position you started in.

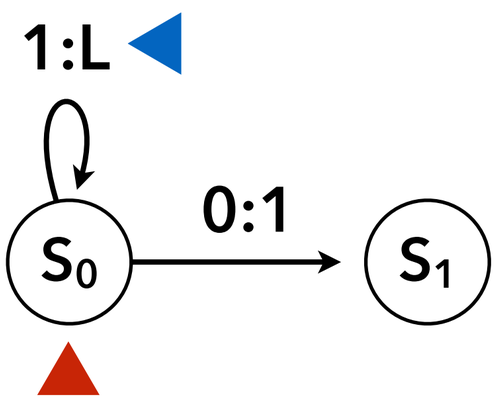
Note: if the final block is empty, then Condition 3 is satisfied by halting anywhere.

Example: +1 T

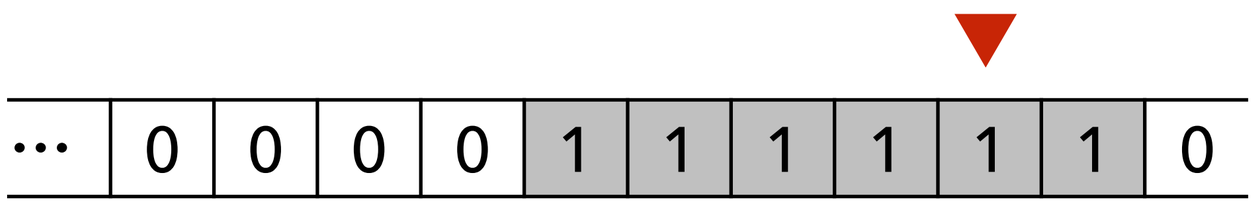


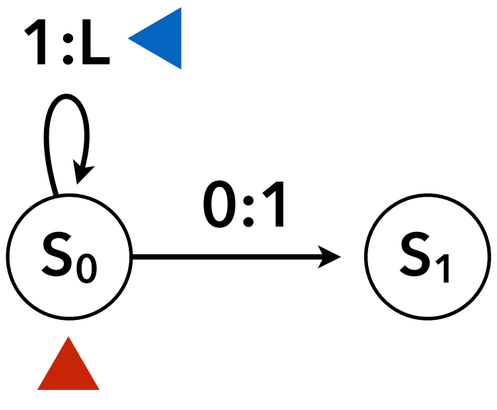
Start with S_0 in the machine, and standard position on the tape.



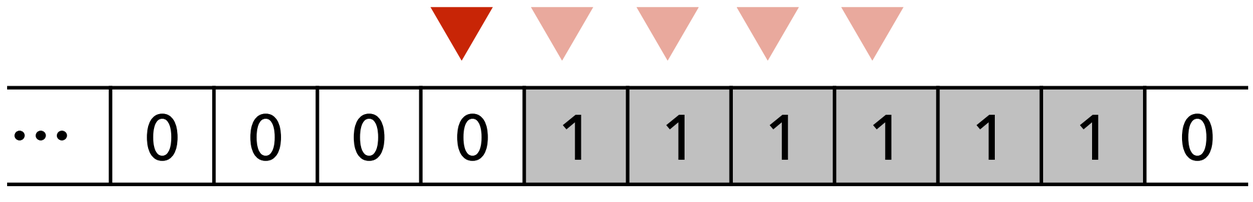


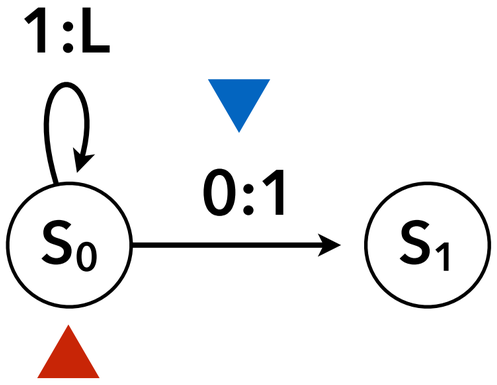
If you read a 1, move left one cell.



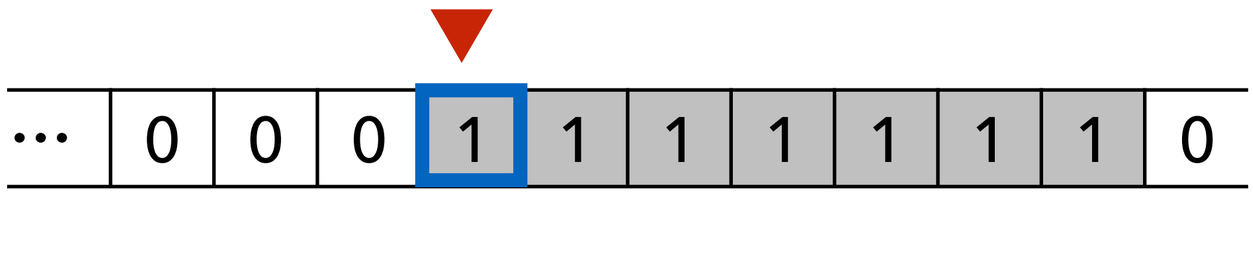


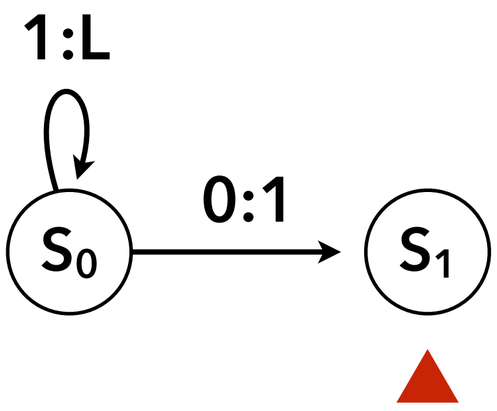
This operation repeats five more times….





If you read a 0, write a 1.





There are no conditions for exiting S_1, so the machine halts.

